

Visualising Forest Sound: Justice-led Ecoacoustic Data Interaction

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ABSTRACT

Recent advances in technology and reduction in hardware costs have made the collection of acoustic environmental recordings more accessible, thereby facilitating easier acquisition of biodiversity data for monitoring, research and conservation. Despite these advances, and the growing power of machine learning and automated techniques to process acoustic data, there exists a lack of tools to facilitate accessible visualisation and exploration of long-duration acoustic recordings collected in the field. Additionally, the exclusion of forest communities and traditional beliefs and practices in tropical forest conservation, originating from colonialism and continuing through neocolonial approaches, often results in tension between conservation practitioners, projects, and forest communities. While there has been a proliferation of citizen science projects, tools and applications arising from ecoacoustics research, there is a need to respond to calls, from other citizen science domains, for this work to engage with “citizens” outside the role of ‘data collector’, towards more co-creative citizen science. Through the lens of participatory design, my research attends to both issues by developing alternative methods of data visualisation that enable more accessible analysis and exploration of big ecoacoustics data, while attending to questions of conservation data justice and participation in forest fringe settings.

CCS CONCEPTS

• **Human-centered computing** → **Visual analytics**; **Ethnographic studies**; **Participatory design**.

KEYWORDS

Ecoacoustics; Citizen Science; Conservation Data Justice; t-SNE

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1 MOTIVATION

Tropical forests will play a major role in shaping the outcomes of both the climate and biodiversity crises, making monitoring an essential practice in revealing long-term trends, contributing to

new knowledge, and informing environmental planning and policy [26]. Yet, there often exist many data-driven barriers to effective decision-making, especially on the continent of Africa, due to insufficient resources or capacity for the collection and analysis of good quality data [3]. Within the toolkit of biodiversity monitoring methods, passive acoustic monitoring (PAM), and the related eco- and bioacoustic fields, combine biological, acoustic and computer sciences, using sound technologies to record, store and analyse acoustic environmental recordings, and “play an increasingly important role in monitoring the biodiversity of terrestrial ecosystem” [38]. However, the field of ecoacoustics suffers from two key challenges detailed as follows:

Visualising and Analysing Big Acoustic Data The increased affordability of hardware has made for greater accessibility in the “collection of acoustic environmental recordings” [29], with sensors being deployed in the field for weeks and months at a time. Advances in machine learning have led to improved processing capabilities in ecoacoustics with researchers often applying supervised deep learning (DL) models, such as Convolutional Neural Networks (CNNs) [6] to analyse short-duration spectrograms (image representations of sound) of lengths between 1 and 30 seconds. These methods have enabled the identification of many species, primarily birds, but continue to be resource and time-expensive, with model training requiring researchers to individually listen to, identify and annotate calls of the target species in hundreds or thousands of spectrograms; much of the collected data is left unexplored by researchers, remaining “opaque and impenetrable” [38].

Citizen Science, Participation and Local Ecological Knowledge Research in ecoacoustics has a “long tradition” of engaging ‘non-experts’ [21] and has been significantly supported by the expertise and labour of enthusiasts, such as birders [25, 37, 43, 45], within ‘contributory’ frames that consist primarily of data collection or labelling. Yet, as with many other conservation technologies, there is an asymmetrical distribution of resources and expertise between the Global North and South. In the African context, there is a “paucity of bioacoustics literature” [3], and the integration of the local ecological knowledge (LEK) of African forest communities remains largely understudied [3]. This lack of engagement with local forest communities [40] or promotion of community agency extends to the development of new technologies that are embedded in forest landscapes. Local communities are “seldom...involved in the technology design of conservation tools which results in the design of non-intuitive systems” [28], often leading to mistrust in the technology itself [27]. There are calls for alternative, “co-creative” approaches to ecoacoustic research that encourage participation and cross-cultural knowledge exchange across the research cycle

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from design, implementation, collection, analysis and interpretation [2, 24, 30].

2 BACKGROUND AND OBJECTIVES

My doctoral research is informed by theories of Conservation Data Justice, follows methodologies of Participatory Design and sits within the technical fields of Interactive Visual Analytics and Sustainable HCI.

With regards to big ecoacoustic data visualisation, in 2014 Towsey et al., developed Long-Duration False Colour (LDFC) Spectrograms, allowing for a full 24 hours of acoustic recording to be visualised by compressing the time component of the frequency-time graph (spectrogram) so that a pixel represented a minute rather than a second or millisecond. More recent work has built on the method of Towsey et al., 2014, applying machine learning models to identify endangered, vulnerable and cryptic species within these long-duration visualisations [14, 39, 46, 47].

However, while the application of these automatic analysis methods is an important step in the understanding of long-duration data analysis, there are significant research gaps. Images of long-duration data, while allowing for the manual inspection of long-term trends and patterns, lack interactivity. The user is unable to explore the data at different scales or in different modes. And, whilst researchers have been able to build detectors for target species and identify them within the LDFC spectrograms, these methods still require time-consuming manual inspection and expert annotation to identify regions or calls of interest [29]. There exists an opportunity to design tools that “abstract and provide insight” into large datasets [11] and facilitate ‘bottom-up hypothesis’ creation [33] to help ecologists and non-experts in the initial exploration and mining of insights from long-duration ecoacoustic data.

Participatory or collective sensing is an emerging genre of research within Sustainable HCI [15], an extension of the HCI field, where interfaces, tools and interactive systems are built, with a focus on “promoting social good and sustainability” [19]. There is a small but growing collection of works [9, 12, 13, 27, 28, 35] extending the “democratic potential of involving end users in data collection”, within participatory sensing [15], towards co-design methods that facilitate connection “through and to” [12] the natural world for biodiversity conservation within remote communities. Some such work [12, 35] at the intersection of participatory design (PD) and acoustic data visualisation has begun to explore alternative modes of interaction through probes such as ‘Ambient Birdhouses’. These probes engage participants through a mixture of false-colour spectrogram annotation, species identification games, trivia questions, and species profiles. The aim of these methods is to “shift the foci from tasking people to gather data to enhancing ways for people to learn and share within the community”.

My research looks to build on these investigations through the co-design of an interactive tool with a forest fringe community in Ghana, a country that is recognised as one of the most advanced tropical African countries in terms of established forest policy and management [5] yet experienced the highest deforestation rate of any tropical country, bar Cote d’Ivoire, between 2017 and 2018. My work asks, as a central research question, *whether interactive visualisations of tropical forest soundscapes can facilitate the*

incorporation of local ecological knowledge (LEK) in ecoacoustic monitoring, and data analysis and interpretation practices?. This question will be probed by attending to the following research objectives:

- (1) To investigate approaches to participatory and justice-led ecoacoustic research, that foster trust, transparency, and agency in community-forest-technology interactions.
- (2) To use machine learning techniques to visualise long-duration ecoacoustic data in different scales and modes.
- (3) To investigate how participatory design can provide insight into community conceptions of sound, forest health and conservation data justice in order to develop visualisation design features that enable community learning, knowledge production and knowledge exchange.

3 RESEARCH METHODS AND PROJECTS

My doctoral research consists of three work packages, responding to each research objective and forming the three main chapters of my thesis. The first project area was conducted in the first year of research in the first field season. The second project area is currently underway and the third will be conducted during and following the second field season.

3.1 Role Play and Gamification: Building Trust and Transparency in Digital Forest Technologies

The first field season was conducted for three months between March and June 2022, in a forest fringe community in the Bosomtwe Range Forest Reserve (BRFR), a Key Biodiversity Area (KBA), in the Ashanti Region of Ghana. Field research was supported by Priscilla Osei, Joseph Kankam, Abena Fosuaa Acheampong, and Ben Ossom from the Kwame Nkrumah University of Science and Technology (KNUST).

After following traditional community entry protocol, consent processes and research presentations, semi-structured interviews were conducted with a total of 15 community members (8 men and 7 women) to understand community members’ local ecological knowledge and perspectives on conservation and the acoustic recorders. A broad range of perspectives was sought, with participants chosen purposively across gender and age to ensure as wide and diverse a set of responses as possible. Despite the standard ethics and consent protocols being followed, my methods were failing to serve the distinct needs and forms of interaction demanded by the community members in this locality. Work in participatory sensing sees ‘citizen scientists’ as not technical, but domain experts, able to use their lived or located experience to better collect data using technological devices [15], but these views assume some level of technological proficiency. Few works, most significantly that of the UCL ExCiteS group [36] have used participatory design to build tools for rural communities, as in my case, with little prior technological interaction. Community members made clear their need to more deeply understand *how* the technology worked, *what* the outputs would be and to have the technical process *proven* to them. They needed a dynamic consent process that responded to their fears, curiosity and agency before the deployment of any acoustic sensors.

This ‘dynamic consent process’ took the form of two interactive workshops. The first focused on the tangibility of acoustic recording hardware, role-playing data collection and building an understanding of the acoustic data itself and was split into four sections. The first activity, the decomposition of an AudioMoth (an acoustic recording device), made the acoustic device, which until then seemed ominous and threatening, tangible. The second, role play and collective sensor deployment facilitated understanding of how the sensors would be deployed in the acoustic pilot study and handed over sensor deployment to community members who assembled the sensor cases, chose a deployment location and affixed the sensor to a chosen tree. One early source of mistrust was the belief by community members that in order to identify wildlife, you *had* to take pictures. Through listening to and identifying bird calls with no imagery in the third activity, community members felt able to understand the methods being used in the field of bioacoustics. This is consistent with the findings of Isak Herman whose doctoral research found that games “provide participant training and knowledge validation which increase potential trust in audio data [17]”. The final activity saw the visualisation of the audio recordings from the previously deployed sensor as both a waveform and a spectrogram, community members became animated by the existence of a variety of patterns, representing different sources of sound, that could be observed.

The second workshop followed a bioacoustic pilot study in which two acoustic sensors were deployed with 4 community members (3 hunters and 1 farmer) for one week. This workshop focused on demonstrating the four major data analysis phases commonly used in bioacoustics: visualisation, segmentation, pattern matching and classification. Instead of demonstrating each stage with the complex computational process, I chose to employ a low-fidelity, human-centred approach using Microsoft PowerPoint and human interaction. In the segmentation stage, for example, a sample recording with no visualisation was played with the participants guessing how many distinct vocalisations there were. We then looked at the spectrogram of the same recording and discussed whether our guesses had changed, the difference in information obtained by human listening alone in contrast with visualisation and how easy or hard it was to differentiate sounds by ear and sight. In contrast to the periods of tension and resistance that were prevalent in earlier interactions, it became clear that through more involved community engagement, community members were not only more trusting but also more intrigued and interested in being a part of the research. One participant reflected that “if it was not for this device, I wouldn’t have known that some of these species exist. So I am intrigued with the device and will commend the idea”.

The contribution of this work is a set of initial methods that acoustic ecologists, and the wider conservation technology community, can employ in order to conduct just and reflexive research in the field as well as foster transparency and explainability in the technological hardware and techniques being used. This work will be built on in the next field trip, focusing particularly on the ‘experimental design’ phase of bioacoustic projects which sees decisions being made on where acoustic sensors are deployed. Commonly, random sampling or transect lines are used but consideration must be taken for how these methods, employed without community input or agency, might cause tensions.

3.2 Visualising Long-Duration Ecoacoustic Data

To address my second research question, and build on the work of Towsey et al., the second project focuses on the application of unsupervised clustering algorithms, specifically the t-distributed Stochastic Neighbour Embedding (t-SNE), a dimensionality reduction (DR) algorithm, to the long-duration acoustic data as an alternative to LDFC visualisations. t-SNE was developed by (Maaten and Hinton, 2008) [42] as a variation of Stochastic Neighbour embedding (SNE). The algorithm works by converting high-dimensional data into a matrix of “pairwise similarities” and has facilitated the exploration of and interaction with a wide range of data, including the visualisation of papers submitted to a Machine Learning conference [1], the classification of the MNIST handwritten numbers dataset [41], as well as the exploration of acoustic recordings in ‘The Infinite Drum Kit’ and ‘Bird Sounds’ interfaces developed by Google [22, 23].

There are two motivations for the use of clustering techniques in this work. The first lies in the hypothesis that the visual characteristics of clusters will allow for more accessible analysis by a wider range of participants. Secondly, clustering methods provide an alternative mode of interaction with long-duration ecoacoustic data that does not necessarily require expert labelling, processing or programming skills in order to visualise large ecoacoustic datasets. Additionally, the probing of important ecological questions through the development of unsupervised approaches to clustering and data visualisation is expected to grow in the near future [6] and in ecoacoustics some of these questions are:

- (1) How does the cluster composition change at different times of the day and can this be related to the presence/absence of species groups or anthropogenic activity?
- (2) Are there distinctions in cluster visualisations across a degradation gradient (e.g., between old growth and secondary forest)? Do these distinctions agree with current, or highlight unobserved, connections between land type and soundscape? And can cluster visualisations replace or support labour-intensive and expensive manual landscape surveys?
- (3) Can species groups be identified and classified within the produced clusters?

3.3 Community Ecoacoustic Visualisation Interface

The third project attends to the final research question, motivated by the idea that visualisations provide “a common language” between ‘experts’ and ‘non-experts’ [34], allowing for greater, more dynamic, participation in the exploration, analysis and interpretation of ecoacoustic data, in contrast to isolated expert data analysis followed by numerical or written reporting of those results. Here two strands of research will be conducted.

The first strand will see a collaboration with an undergraduate student at Cambridge University, building on the LDFC visualisations. The aim here is to allow users to visualise the spectrograms at different temporal scales to enable more accessible and intelligible data exploration. Two main approaches will be experimented with. Firstly ‘focus+context’ methods [10, 20] will be employed in order to offer richer modes and patterns of interaction (e.g., the full 24-hour spectrogram) and contextual data views (e.g., a specific

half-hour spectrogram within the original LDFC) [16]. Secondly, zooming spectrograms will be developed to allow for data exploration at varying temporal scales, building on current work by Towsey and Trusking 2015 who produced and layered images of long-duration acoustic data at different scales. Their work called for further research to create smoother user zooming between different temporal scales and to build this into an interactive interface [44].

The second strand will focus on creating interactive t-SNE visualisations of the same ecoacoustic data, extending the work of the second project. The same input features will be used as in the LDFC methods, where the acoustic indices of a one-minute sample from a long-duration recording are set as the features for the t-SNE clustering algorithm, to enable later comparison between the two visualisation methods in user studies. There is considerable literature on interactive machine learning and interactive visual analytics, including the interaction with t-SNE and other clustering algorithms [7, 8, 18, 31], but these works are largely aimed at expert interaction. As highlighted in the recent publication of (Birhane et al., 2022), there is a need for participatory practices in AI, but also within the field of visual analytics that caters almost exclusively to ‘highly trained experts’ [32], to engage the most marginalised outside of the paradigm where participation is “conceived as a technical work done by participants for the sake of economic efficiency” [4]. The aim of this project is to understand the “views, needs and perspectives” of community members in relation to visualising, analysing and interpreting ecoacoustic data for decision-making. In this way, the visualisation interface probes two key questions, firstly, *what knowledge is constructed through interaction with a novel ecoacoustic visual interface* and secondly, *how can the flow of knowledge between interface and user become reciprocal, flowing not just from the interface for consumption by the user, but imparted from the user and integrated into the visualisation as well*. An initial prototype will be developed before the next field trip, ready for initial testing with community members as well as design workshops where community members can iterate on, adapt and influence subsequent development on the interface.

Once development on both interactive visualisations is complete, a final set of user studies will be conducted between Cambridge-based conservationists and the community members in Ghana in order to investigate the ease of use, accessibility, facilitation of knowledge exchange/production, perceived benefit and facilitation of creativity and environmental connection of each interface.

4 CONCLUSIONS

This work will have key contributions to the HCI, Participatory Design and (Ecoacoustic) Conservation fields respectively. A set of participatory design methods and implications to be employed by designers and engineers building interactive visualisations to be used in forest fringe settings will be presented. As well as this, a prototype tool and set of methodologies that enable the integration of local ecological knowledge in ecoacoustic research, and the conduct of just, ethical, transparent and dynamic participation with forest fringe communities where conservation technologies like sensors are being deployed, will be produced. The developed interface will

allow researchers and community members to probe ecoacoustic data in an easy-to-use generative and collaborative manner.

5 RESEARCH SITUATION

I am a second-year PhD candidate at the University of Cambridge, based in the Computer Laboratory through the Artificial Intelligence for Environmental Risk (AI4ER) Centre for Doctoral Training (CDT). I am primarily supervised by Professor Alan Blackwell (Professor of Interdisciplinary Design in the Computer Laboratory at Cambridge University) and co-supervised by Professor Jennifer Gabrys (Chair in Media, Culture and Environment in the Department of Sociology at the University of Cambridge), Dr Adham Ashton-Butt (Senior Ecologist at the British Trust for Ornithology (BTO)) and Professor Emmanuel Acheampong (Associate Professor at the Faculty of Renewable and Natural Resources at the Kwame Nkrumah University for Science and Technology (KNUST)). I expect to submit my doctoral thesis in the autumn of 2024.

My research interests have always been interdisciplinary, being especially drawn to the issues and opportunities that arise at the intersection of technology, ecology and community. The HCI field appealed to me as it seemed to be a space where truly interdisciplinary research was not just conducted but widely celebrated. I found interacting with other HCI researchers and literature to be stimulating and generative for my own work in a way that the wider Artificial Intelligence field did not. I enjoy being able to employ both qualitative, quantitative and field methods to observe, highlight and engage with the links that tie the three domain spaces I am working in together. I am relatively new to HCI but I am greatly influenced, motivated and supported by my current work as an environmental justice activist, commentator, organiser and educator, and the technical, programming and mathematical training received from my Undergraduate degree in Astrophysics.

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